

## **MODULE II**

### **SYLLABUS**

- Surveying: Importance, objectives and principles.
- Bricks: - Classification, properties of good bricks, and tests on bricks
- Stones: - Qualities of good stones, types of stones and their uses.
- Cement: - Good qualities of cement, types of cement and their uses.
- Sand: - Classification, qualities of good sand and sieve analysis (basics only).
- Timber: - Characteristics, properties and uses.
- Cement concrete: - Constituent materials, properties and types,
- Steel: - Steel sections and steel reinforcements, types and uses.
- Modern construction materials: - Architectural glass, ceramics, plastics, composite materials, thermal and acoustic insulating materials, decorative panels, waterproofing materials, modern uses of gypsum, pre-fabricated building components (brief discussion only)

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2

### **SURVEYING**

- The art of determining the relative positions of points on above or beneath the surface of the earth by means of measurements in horizontal and vertical plane

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3

## **IMPORTANCE OF SURVEYING**

- 1) The first necessity in surveying is to prepare site plan and section of the area to be covered by the project
  - ❖ From these prepared site plan and sections – the best possible alignment, amount of earth work and other necessary details depending upon on the nature of the project may be calculated
- 2) The planning and design of all civil engineering projects such as railways, highways, tunneling, irrigation, dams, reservoirs, waterworks, sewage works, airfields, ports, massive buildings etc. are based on the surveying measurements

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4

## **IMPORTANCE OF SURVEYING**

- 3) The measurement of land and the fixation of its boundaries cannot be done without surveying
- 4) The economic feasibility of a project cannot be properly ascertained without undertaking a survey work
- 5) The execution of hydrographic charting and mapping requires surveying
- 6) Preparation of topographic map of a land surface of the earth requires surveying

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5

## **OBJECTIVES OF SURVEYING**

- 1) To prepare plan or map of an area or location
- 2) For the proper location of buildings, railway, dams, bridges, tunnels etc.
- 3) To determine the height of objects in a vertical plane
- 4) Measurement of distances between two points
- 5) Determination of relative positions of points
- 6) Establishment of boundaries of properties with reference to available records
- 7) Applications in Geographic Information System (GIS)

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6

# PRINCIPLES OF SURVEYING

➤ Surveying has the following two main principles

- 1) Working from whole to the part
- 2) Location of a point by measurement from two points of reference

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7

# PRINCIPLES OF SURVEYING

➤ Working from whole to part

- ❖ Main principle of surveying
- ❖ In surveying large areas, a system of primary control points are identified and they are located with high precision
- ❖ In between primary control stations minor control points are established with less precision methods
- ❖ With the help of minor control points, details are surveyed by adopting any methods of surveying

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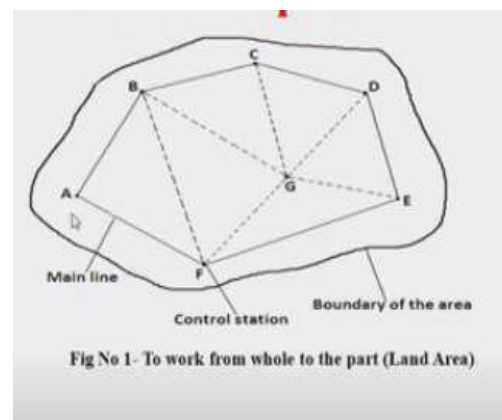
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# PRINCIPLES OF SURVEYING

**The whole area is first enclosed by main stations (controlling stations) and main survey lines (controlling lines).**

**The area is then divided into a number of parts by forming well conditioned triangles.**

**The main survey lines are Measured very accurately with a Standard chain and then the sides of triangles are measured.**



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9

# PRINCIPLES OF SURVEYING

## ➤ Working from whole to part – advantages

- ❖ If any error is made in the measurement, it will affect only that particular measurement, therefore it will not affect the accuracy of whole survey
- ❖ Adopted to prevent accumulation of error and to localize the minor errors in the frame work of control points
- ❖ If not, the error would expand to greater magnitude if reverse process is followed, thus making the work uncontrollable

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10

# PRINCIPLES OF SURVEYING

## ➤ Location of a point by measurement from two points of reference

- ❖ The relative positions of points to be surveyed should be located by measurements from at least two points of reference, the positions of which have already been fixed
- ❖ Two points are selected in the field and the distance b/w them is measured and plotted on paper
- ❖ With respect to these control points any other points in the field may be located either by linear and angular measurements

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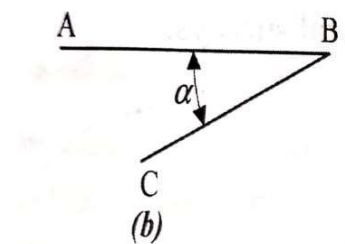
11

# PRINCIPLES OF SURVEYING

## ➤ Location of a point by measurement from two points of reference

- ❖ Let A and B be 2 control points and the position of C is to be plotted by any of the following methods

a) By measuring distance BC and angle  $\alpha$

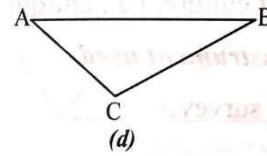
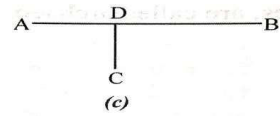


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12

# PRINCIPLES OF SURVEYING

- b) By dropping a perpendicular from C and measuring either AD and CD or BD and CD
- c) By measuring distances AC and BC

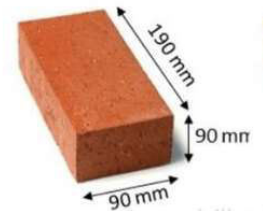


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13

## BRICKS

- Made of soil
- Used for construction of buildings, bridges etc
- Size of brick
  - Standard size – 19 x 9 x 9 cm
  - Nominal size – 20 x 10 x 10 cm



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## BRICKS



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# COMPOSITION

- Silica : 50 – 60% (prevents cracking, shrinking, warping)
- Alumina : 20 – 30% (impart plasticity)
- Lime : 5% ( prevents shrinking)
- Oxide of iron : 5 – 6 % (impart red colour)
- Magnesia : small amount (imparts yellow colour and prevents shrinking)

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16

## CLASSIFICATION

- Bricks are classified as
  1. On the basis of Compressive strength
  2. On the basis of manufacture
    - 1) Unburned/sundried bricks
    - 2) Burned bricks

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17

## CLASSIFICATION

1. On the basis of Compressive strength

| Class | Average compressive strength<br>not less than(N/mm <sup>2</sup> ) |
|-------|-------------------------------------------------------------------|
| 350   | 35                                                                |
| 300   | 30                                                                |
| 250   | 25                                                                |
| 200   | 20                                                                |
| 175   | 17.5                                                              |
| 150   | 15                                                                |
| 125   | 12.5                                                              |
| 100   | 10                                                                |
| 75    | 7.5                                                               |
| 50    | 5                                                                 |
| 35    | 3.5                                                               |

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18

## CLASSIFICATION

### 2. On the basis of manufacture

- 1) Unburned/sundried bricks
  - ❖ used only for temporary structures
- 2) Burned
  - ❖ used for permanent construction
  - ❖ Classified into 4 categories
    - a) First class brick
    - b) Second class brick
    - c) Third class brick
    - d) Kiln rejects

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19

## CLASSIFICATION



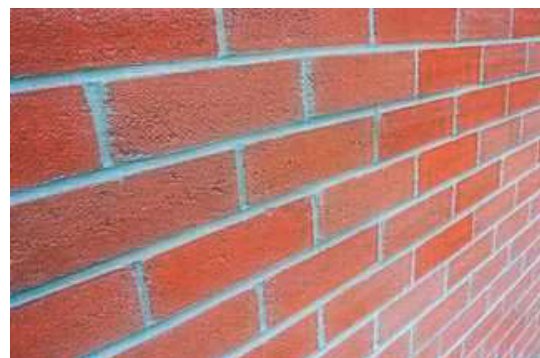
### a) First class brick

- ❖ Machine or table moulded and burnt in kilns
- ❖ Rectangular shape with sharp edges and uniform size
- ❖ Clean and even surface and free from cracks, flaws or lumps
- ❖ Well burnt and shall not be scratchable by finger nails
- ❖ The crushing strength shall not be less than  $10.5\text{N/mm}^2$
- ❖ Water absorption shall not exceed 15% of dry weight when immersed in water for 24 hours
- ❖ Used for construction of load bearing walls, pavements, walkways, flooring etc

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20

## CLASSIFICATION



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## CLASSIFICATION



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## CLASSIFICATION

### b) Second class brick

- ❖ Ground moulded and burnt in kilns
- ❖ Rectangular in shape with well defined and sharp edges but slightly irregular in size
- ❖ Shall be perfect and free from cracks
- ❖ Water absorption shall not exceed 22% of dry weight when immersed in water for 24 hours
- ❖ The crushing strength shall not be less than  $7 \text{ N/mm}^2$
- ❖ Used for masonry construction where faces are to be plastered



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23

## CLASSIFICATION

### c) Third class brick

- ❖ Ground moulded and burnt in clamp
- ❖ May be slightly over burnt or under burnt
- ❖ May not have regular shape and size
- ❖ Produce dull sound when struck together
- ❖ Water absorption shall not exceed 24% of dry weight when immersed in water for 24 hours
- ❖ Used for construction of unimportant structures, and in structures where load bearing capacity is less such as sheds etc



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24



## CLASSIFICATION



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25

## CLASSIFICATION

### d) Kiln rejects

- ❖ Ground moulded and burnt in clamp
- ❖ Over burnt with irregular shape and size
- ❖ Dark colour and brittle
- ❖ Not used for normal engineering works, may be for road construction and as aggregate for concrete



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26

## PROPERTIES OF GOOD BRICKS

- ❖ Should have perfect edges, well burnt in kilns, copper coloured, free from cracks with sharp and square edges
- ❖ Uniform in shape and standard in size
- ❖ When broken should show a bright homogenous and uniform compact structure free from voids
- ❖ Should produce clear metallic ringing sound when struck with each other
- ❖ Should not break when dropped from a height of 1 m
- ❖ Should be sufficiently hard i.e, no nail impression must be present when scratched

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27

# PROPERTIES OF GOOD BRICKS

- ❖ Water absorption should be not greater than 15% for first class bricks & 22% for second class brick & 24% for third class brick when soaked in water for 24 hours
- ❖ Should have low thermal conductivity and shall be sound proof
- ❖ Should have a fire protection of 6 hours
- ❖ Should not show any white or grey deposits when immersed in water and dried i.e, efflorescence
- ❖ Should not have a crushing strength below  $3.5 \text{ N/mm}^2$

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28

# TEST ON BRICKS

- ❖ The following tests are conducted to find out various properties of bricks
  1. Crushing or compressive strength of brick
  2. Water absorption test
  3. Structure
  4. Hardness test
  5. Soundness test
  6. Shape and size test
  7. Efflorescence test
  8. Toughness test

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29

# TEST ON BRICKS

## ***1. Crushing or compressive strength of brick***

- Crushing strength of bricks is determined by placing brick in compression testing machine.
- After placing the brick in compression testing machine, apply load on it until brick breaks.
- Note down the value of failure load and find out the crushing strength value of brick. Minimum crushing strength of brick is  $3.50 \text{ N/mm}^2$ .
- If it is less than  $3.50 \text{ N/mm}^2$ , then it is not useful for construction purpose

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30

# TEST ON BRICKS



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31

# TEST ON BRICKS

## 2. Water absorption test

- ❖ Used to find out the amount of water absorbed by the brick
- ❖ Dry weight of brick is noted, and the it is kept immersed in water for 24 hours
- ❖ It is taken out, wiped with cloth and weighed
- ❖ Water absorption (%) =  $\frac{\text{increase in weight of brick}}{\text{original weight of brick}} \times 100$
- ❖ Should not be more than 20% for good specimen and not more than 15% for Class I bricks

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32

# TEST ON BRICKS

## 3. Structure

- ❖ When brick is broken, it should be homogenous, compact and free from holes and lumps

## 4. Hardness test

- ❖ Scratch on the brick by fingernails. If no impressions are left on it, then brick is hard

## 5. Soundness test

- ❖ When two bricks are knocked against each other it should give metallic ringing sound and should not be broken

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33

# TEST ON BRICKS

## 6. Shape and size test

- ❖ To check shape and size, 20 bricks are selected at random and stacked length wise
- ❖ Total length is then measured
- ❖ Similarly, measurements are taken after stacking them along width and height
- ❖ for brick size 190 x 90 x 90 mm, the IS code permits the following values
  - ✓ Length – 3680 to 3920 mm
  - ✓ Width – 1740 to 1860 mm
  - ✓ Height – 1740 to 1860 mm

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34

# TEST ON BRICKS

## 7. Efflorescence test

- ❖ Used to test the presence of soluble salts
- ❖ Brick is immersed in water for 24 hours
- ❖ Then it is taken out and dried in shade
- ❖ After drying if a thin layer of white or grey patches are observed on the surface indicates the presence of harmful soluble salts or alkalis

## 8. Toughness test

- ❖ A brick is dropped flat from a height of 1 meter on a firm platform it should not break, then the brick is tough enough

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35

# ASSIGNMENT I

Timber: - Classification, properties and uses

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36

# STONES

- ❖ A '*naturally available building material*' available in the form of rocks, which is cut to required size and shape and used as a building block
- ❖ In ancient era, out of the three basic materials available for construction, viz., wood, stone, and mud or clay, stone was the principal construction material
- ❖ Used as the structural material for construction of walls, columns, arches, interior and exterior finishing material, flooring and roofing material
- ❖ Also abundantly used for the construction of retaining walls, roads, paths, steps etc.
- ❖ Examples – Taj Mahal, Red Fort, Jama Masjid

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37



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# QUALITIES OF GOOD STONES

❖ The following are the important characteristics of good building stones

- |                      |                                    |
|----------------------|------------------------------------|
| 1. Crushing strength | 8. Porosity and absorption         |
| 2. Hardness          | 9. Durability & Weather resistance |
| 3. Specific gravity  | 10. Cost                           |
| 4. Structure         | 11. Seasoning                      |
| 5. Texture           | 12. Resistance to fire             |
| 6. Appearance        | 13. Workability                    |
| 7. Percentage wear   | 14. Toughness                      |

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40

## STONES

### 1. Crushing strength

- ❖ The load required per unit area to crush a specimen of stone under compressive load ( $\text{N/mm}^2$ )
- ❖ IS code recommends a minimum crushing strength of  $3.5 \text{ N/mm}^2$  for any building block.

### 2. Hardness

- ❖ Important property to be considered when stone is used for flooring and pavement
- ❖ For road work, the coefficient of hardness  $> 17$
- ❖ For buildings, the coefficient of hardness  $> 14$

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41

## STONES

### 3. Specific gravity

- ❖ Denser stones are stronger and lighter ones are weaker
- ❖ Specific gravity  $< 2.4$  are considered unsuitable for buildings

### 4. Structure

- ❖ Manner in which particles forming the mass of stones are arranged.
  1. Stratified – construction of walls, columns etc.
  2. Un-stratified – foundation works, dams, retaining walls etc.
  3. Foliated – not strong, not used for masonry works

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42



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43

## STONES

### 5. Texture

- ❖ Arrangement, size and shape of grains in stones
- ❖ Fine grained stones with homogenous distribution looks attractive and used for carving

### 6. Appearance

- ❖ Compact grains with uniform and attractive colour is durable
- ❖ Marble and granite when polished get very good appearance – used for face works in buildings

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44

## STONES

### 7. Percentage wear

- ❖ Important property to be considered in selecting aggregate for road works and railway ballast
- ❖ Wear of a good stone < 2%

### 8. Porosity

- ❖ All stones have pores and absorb water
- ❖ The reaction of water with minerals of stones cause disintegration – less durable
- ❖ Water absorption - % of water absorbed by stone when immersed in water for 24 hours and should be < 5

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45



# STONES

## 9. Durability & Weather resistance

- ❖ Good stone should be durable and weather resistant
- ❖ Stones which are crystalline in structure, homogenous, and close grained have better durability
- ❖ Stones selected should have good resistance against adverse effects of natural forces like wind, rain and heat

## 10. Cost

- ❖ Proximity of quarry to building site brings down the cost of transportation thereby, reducing overall cost

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46

# STONES

## 11. Seasoning

- ❖ The process of removing moisture from pores
- ❖ Strength of stone improves if its moisture is removed before using it
- ❖ The best way of seasoning is to allow it to the action of nature (exposing to atmosphere) for 6 to 12 months

## 12. Resistance to fire

- ❖ Depends on its mineral composition
- ❖ Sand stones resist fire better
- ❖ Argillaceous materials, though poor in strength has good fire resistance

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47

# STONES

## 13. Workability

- ❖ Should be workable so that cutting, bringing it to the required shape and size are economical

## 14. Toughness

- ❖ Resistance to impact
- ❖ Tough enough to sustain stresses developed due to vibrations
- ❖ For road works toughness index >19
- ❖ Toughness index b/w 13 to 19 - medium tough
- ❖ Toughness index < 13 – poor stones

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48

# TYPES OF STONES

➤ Rocks are classified under four heads

## 1. Geological classification

- a) Igneous
- b) Sedimentary
- c) Metamorphic

## 2. Physical classification

- a) Stratified
- b) Un-stratified
- c) Foliated

## 3. Chemical classification

- a) Silicious
- b) Argillaceous
- c) Calcareous

## 4. Hardness classification

- a) Soft
- b) Medium
- c) Hard
- d) Very hard

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49

# TYPES OF STONES

## 1. Geological classification – based on origin

### a) Igneous rocks

- ❖ Formed by cooling of hot molten mass – *magma* from the heart of the earth
- ❖ Granite, trap and basalt.

### b) Sedimentary rocks

- ❖ Formed by gradual deposition or sedimentation under water due to weathering action of water, wind and frost on existing rocks on the surface of earth.
- ❖ Sandstone, limestone, mudstone etc.

### c) Metamorphic rocks

- ❖ Formed due to change in the character of pre-existing rocks due to metamorphic action of heat and pressure.
- ❖ Limestone and marl becomes marble
- ❖ Mudstone becomes slate
- ❖ Sandstone becomes quartzite

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50

# TYPES OF STONES

## 2. Physical classification – based on structure

### a) Stratified rocks

- ❖ Have layered structure

### b) Un-stratified rocks

- ❖ Do not have distinct plane of stratification

### c) Foliated rocks

- ❖ Have a tendency to split along a definite direction only

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51

# TYPES OF STONES

## 3. Chemical classification

### a) Silicious rocks

- ❖ Principal constituent is silica
- ❖ Granite, trap, sandstones etc.

### b) Argillaceous rocks

- ❖ Principal constituent is argil i.e, clay
- ❖ Slates, laterites etc.

### c) Calcareous rocks

- ❖ Principal constituent is calcium carbonate
- ❖ Limestone, marble

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52

# TYPES OF STONES

## 4. Hardness classification

### a) Soft rocks

- ❖ Examples – talc, gypsum, slate, sand stone

### b) Medium rocks

- ❖ Examples – lime stone and dolomite

### c) Hard rocks

- ❖ Examples – deccan trap, basalt, quartzite

### d) Very hard rocks

- ❖ Examples – granite, trap rock

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53

# COMMON BUILDING STONES

## 1. Basalt and trap

- ❖ Igneous rocks
- ❖ Hard, compact, durable, un-stratified stone
- ❖ Used for rubble masonry works in bridge piers, river walls and dams, as decorative features in structures



## 2. Granite

- ❖ Igneous rocks
- ❖ Hard, compact, un-stratified stone
- ❖ Used for heavy construction such as dams, bridges, sea walls, for flooring



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54

# COMMON BUILDING STONES

## 3. Sand stone

- ❖ Sedimentary rocks
- ❖ Stratified
- ❖ Used for building construction, as aggregate in concrete



## 4. Lime stone

- ❖ Sedimentary rocks
- ❖ Available in grey, blue and brown colour
- ❖ Used for regular building construction, paving, flooring, for manufacturing of lime and cement



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55

# COMMON BUILDING STONES

## 5. Kankar

- ❖ Sedimentary rocks
- ❖ Impure limestone
- ❖ Used as building material, railway ballast



## 6. Slate

- ❖ Metamorphic rocks
- ❖ Compact, fine, hard and durable
- ❖ Used as roofing tiles, slabs, pavements etc.



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56

# COMMON BUILDING STONES

## 7. Laterite

- ❖ Metamorphic rocks
- ❖ Has porous or cellular structure
- ❖ When used as building material its surface should be plastered



## 8. Marble

- ❖ Metamorphic rocks
- ❖ Available in different pleasing colours
- ❖ Used for facing and ornamental works, columns, flooring etc



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57

# USES OF STONES

- Stones are used in following civil engineering constructions
  - ❖ For the construction of foundations, walls, columns, arches
  - ❖ For flooring
  - ❖ Used as damp proof courses, lintels, roofing materials
  - ❖ Stones with good appearance are used for face works of buildings
  - ❖ For paving of roads, footpaths and open space round the buildings
  - ❖ In the construction of piers and abutments of bridges, dams

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58

# USES OF STONES

- Stones are used in following civil engineering constructions
  - ❖ Crushed stones are used as
    - a) Base course for road
    - b) As a basic inert material in concrete
    - c) For making artificial stones and building blocks
    - d) As railway ballast
    - e) When mixed with tar they form finishing coat

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59

## SAND

- An important building material
- Formed by decomposition of rocks
- Consists of small grains of silica ( $\text{SiO}_2$ )
- Natural sources of sand
  1. Pit sand – excavated from a depth of about 1-2m ground level
  2. River sand – obtained from banks or beds of rivers
  3. Sea Sand – obtained from sea shores

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60

## **SAND - CLASSIFICATION**

### **1. Fine Sand**

- The sand passing through a sieve with clear opening of 1.5875 mm
- Used for plastering

### **2. Coarse Sand**

- The sand passing through a sieve with clear opening of 3.175 mm
- Used for masonry work

### **3. Gravelly Sand**

- The sand passing through a sieve with clear opening of 7.62 mm
- Used for concrete work

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61

## **QUALITIES OF GOOD SAND**

- Sand should be clean, coarse and chemically inert.
- Sand should be free from any organic or vegetable matter.
- It should contain sharp, angular, coarse and durable grains.
- Sand should not contain salts which attract moisture from atmosphere.
- Sand should be well graded i.e, it should contain particles of various sizes in suitable proportions.

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62

## **USES OF SAND**

- Used in concrete and for making mortars in building construction.
- Used in building construction for filling underground floors and filling behind retaining walls.



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63

## SAND - SIEVE ANALYSIS

- Aggregates are inert materials basically used as filler with binding materials in the production of mortar and concrete
- According to size aggregate are classified as



1. Fine Aggregate
  - Most of the particles pass through 4.75 mm IS sieve
  - Eg. sand
  - Helps to increase the volume of concrete
2. Coarse aggregate
  - Most of the particles are retained in the 4.75 mm IS sieve
  - Eg. Crushed stones, natural gravel
  - Act as main load-bearing component of the concrete



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64

## SAND - SIEVE ANALYSIS

- The particle size distribution of an aggregate is determined by *Sieve Analysis* is termed as **grading of aggregate** or **gradation**.
- If all particles are of uniform size the compacted mass will contain more voids, whereas an aggregate comprising of particles of various sizes will give a mass containing lesser voids.
- Grading pattern is assessed by sieving a sample successively through all the sieves mounted one over the other in order of size, with larger sieve on top
- The series of sieves are set in the order,
  - Coarse aggregate – 80mm, 40mm, 20mm, 10mm, 4.75mm
  - Fine aggregate – 2.36mm, 1.18mm, 600µm, 300 µm, 150µm



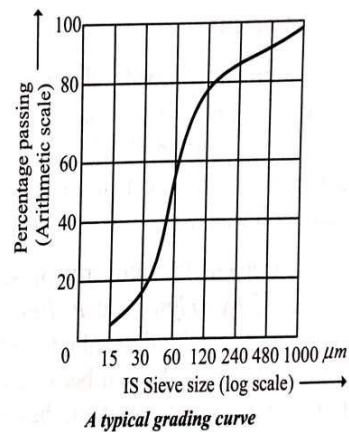
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65



## SAND - SIEVE ANALYSIS

- Grading is expressed in terms of percentage by weight retained on a series of sieves taken in order
- 3 categories of aggregate grading
  - Well graded
  - Gap graded
  - Poorly graded

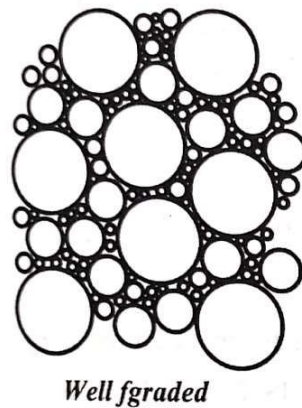


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67

## SAND - SIEVE ANALYSIS

- Well graded
  - Contains particles of a wide range of sizes and has a good representation of all sizes.
  - Desirable for making concrete, as the space between larger particles is effectively filled by smaller particles to produce a well-packed structure.
  - Characterized by the *S-shaped gradation curve*.

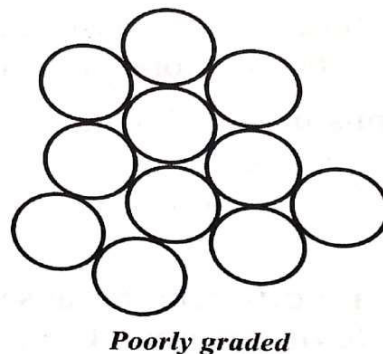


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68

## SAND - SIEVE ANALYSIS

- Poor graded
  - Characterized by small variation in size.
  - Contains aggregate particle size that are almost of the same size.
  - Aggregates are not effectively packed, creating relatively large voids in concrete.
  - Generally requires excessive amounts of cement pate to fill the voids making them uneconomical.



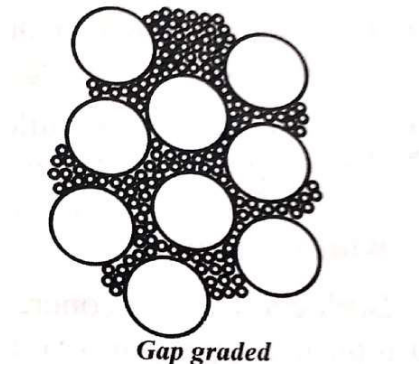
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69

## SAND - SIEVE ANALYSIS

### c) Gap graded

- Contains aggregate particles in which one or more intermediate size particles are missing.
- Resulting concrete is very stiff and has low workability.
- Gap graded concretes fall in between well-graded and poorly graded in terms of performance and economy.



01-07-2023

70

## CEMENT

- ❖ Product obtained by burning a well proportioned mixture of Siliceous, Argillaceous and Calcareous materials and crushing the same into a fine powder
- ❖ A binder substance that sets and hardens independently, and can bind other materials together
- ❖ Bogue's compounds
  - ✓ Tricalcium Silicate ( $C_3S$ )
  - ✓ Dicalcium Silicate ( $C_2S$ )
  - ✓ Tricalcium Aluminate ( $C_3A$ )
  - ✓ Tetracalcium Aluminoferrite ( $C_4AF$ )

01-07-2023

71

## PROPERTIES OF CEMENT

### ➤ PHYSICAL PROPERTIES OF CEMENT

- a) Fineness of cement
- b) Soundness of cement
- c) Initial and final setting time
- d) Standard consistency
- e) Compressive strength
- f) colour

### ➤ CHEMICAL PROPERTIES

01-07-2023

72

# PROPERTIES OF CEMENT

## a) Fineness of cement

- ❖ Measure of the size of the cement particles
- ❖ Rate of the chemical reaction is largely dependent on the fineness
- ❖ Fineness obtained by the sieve test is expressed in %
- ❖ Finer the cement, greater is the reaction
- ❖ For good quality cement, it should be  $< 10\%$

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73

# PROPERTIES OF CEMENT

## b) Soundness of cement

- ❖ Measure of resistance against large volumetric change after setting
- ❖ Free lime and magnesia present in excess causes volume change, causing cracks
- ❖ Measured by *Le Chatlier Test* and expressed in mm
- ❖ For ordinary cement, it should be  $< 10$  mm

01-07-2023

74

# PROPERTIES OF CEMENT

## c) Initial and final setting time

- ❖ The process of changing cement paste to hard solid mass – setting of cement
- ❖ Time taken for this setting – setting time of cement
- ❖ The time at which cement paste starts losing its plasticity after the addition of water – Initial setting time - should be  $< 30$  minutes (OPC)
- ❖ The time elapsed b/w the moment at which water is added to the cement, and when the paste has completely lost its plasticity – Final setting time - should be  $< 10$  hours (OPC)
- ❖ Measured by Vicat apparatus

01-07-2023

75

# PROPERTIES OF CEMENT

## d) Standard Consistency

- ❖ The quantity of water required to produce a workable cement paste
- ❖ Found out by Vicat apparatus
- ❖ Normally this value ranges from 25% to 35% of dry weight of cement

01-07-2023

76

# PROPERTIES OF CEMENT

## e) Compressive strength

- ❖ Defined as the characteristic compressive strength at 28 days of curing of a cement mortar cube of 50 cm<sup>2</sup> face area prepared in the ratio of 1:3 (*1 part of cement and 3 parts of standard sand*) and  $(P/4 + 3)\%$  of water
- ❖  $P$  – standard consistency
- ❖ Testing is conducted using Standard Compressive Testing Machine
- ❖ For OPC, the compressive stress  $> 33 \text{ N/mm}^2$  at 28 days of curing

01-07-2023

77

# PROPERTIES OF CEMENT

## f) Colour

- ❖ Cement should be of uniform colour of grey with light greenish shade
- ❖ It should be free from lumps
- ❖ Should be cool when felt with hand

01-07-2023

78

# PROPERTIES OF CEMENT

## ➤ CHEMICAL PROPERTIES OF CEMENT

- ❖ Ratio of percentage of alumina to that of iron oxide  $> 0.66$
- ❖ Ratio of percentage of lime to that of alumina, iron oxide, silica  $> 0.66$  and  $< 1.02$
- ❖ Sulphur content as  $\text{SO}_3 < 2.5\%$
- ❖ Weight of insoluble residue  $< 4\%$
- ❖ Weight of magnesia  $< 6\%$

01-07-2023

79

# TYPES OF CEMENT

- |                              |                               |
|------------------------------|-------------------------------|
| 1) Ordinary Portland Cement  | 6) Coloured cement            |
| 2) Rapid hardening cement    | 7) Quick setting cement       |
| 3) Low heat cement           | 8) Expansive cement           |
| 4) Portland Pozzolana Cement | 9) Acid resisting cement      |
| 5) Portland slag cement      | 10) Sulphate resisting cement |

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80

# TYPES OF CEMENT

## 1. Ordinary Portland Cement

- ❖ Available in market in three different grades 33 (IS.269:1989), 43 (IS 8112:1989) and 53 (IS 12269: 1987)
- ❖ 33, 43, and 53 represents the cube strength of cement mortar (1:3) cubes after 28 days of curing
- ❖ Uses
  - ☐ Cement mortar
  - ☐ Slab, column and concrete hollow brick
  - ☐ Pointing and plastering
  - ☐ Building material in brick and stone masonry

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81

# TYPES OF CEMENT

## 2. Rapid Hardening Cement (IS 8041:1990)

- ❖ Develops strength rapidly
- ❖ Strength at the age of 3 days is same as that expected of OPC at 7 days
- ❖ Uses
  - ☐ In prefabricated constructions
  - ☐ Road repair works
  - ☐ Situations where the formwork is required to be removed early
  - ☐ In cold weather climate for better resistance against frost damage
  - ☐ Short period of curing

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82

# TYPES OF CEMENT

## 3. Low heat Cement (IS 12600:1989)

- ❖ The considerable amount of heat liberated during hydration of cement may lead to harmful effects like serious cracks in case of mass concreting such as dams, bridge abutments, retaining walls etc.
- ❖ Therefore it is necessary to limit the rate of heat evolution of the cement used in such structures by reducing  $C_3A$  and  $C_3S$  and increasing the  $C_2S$  content in the cement
- ❖ Uses
  - ☐ Mass concreting like dams, abutments etc.

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83

# TYPES OF CEMENT

## 4. Portland Pozzolana Cement (IS 1489:1991)

- ❖ Made either by inter grinding Portland cement clinker and Pozzolana or by uniform blending of Portland cement and fine Pozzolana
- ❖ Pozzolana may be natural materials like volcanic ash or materials like flyash
- ❖ In India, burnt clay, shale, flyash is used
- ❖ Pozzolana's content is 10 -30% by weight of cement
- ❖ PPC is finer than OPC due to Pozzolanic action – reduces microcracks

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84

# TYPES OF CEMENT

## 4. Portland Pozzolana Cement (IS 1489:1991)

- ❖ This cement is recommended for
  - ☐ Marine structures
  - ☐ Sewers and sewage disposal works
  - ☐ Hydraulic structures like dams, reservoir, canals, aqueducts
  - ☐ Mass structures like bridge piers and thick foundations

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85

# TYPES OF CEMENT

## 5. Portland slag Cement (IS 455:1989)

- ❖ Slag – waste product of steel industry
- ❖ This cement is produced by grinding Portland cement clinkers and granulated blast furnace slag (<65%) with gypsum
- ❖ Advantages
  - ☐ Reduced heat of hydration
  - ☐ Reduced permeability
  - ☐ Better resistance against corrosion of steel reinforcement
  - ☐ Increased resistance to chemical attack

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86

# TYPES OF CEMENT

## 5. Portland slag Cement (IS 455:1989)

- ❖ Uses
  - ☐ For mass concrete works
  - ☐ Constructions in marine environment

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87

# TYPES OF CEMENT

## 6. Coloured Cement

- ❖ Obtained by mixing colouring pigments (5 – 10%) with ordinary cement
  - ✓ Green – chromium Oxide
  - ✓ Blue – Cobalt Brown
  - ✓ Red, Yellow – Iron Oxide
  - ✓ Black – Mangense Oxide

01-07-2023

88

# TYPES OF CEMENT

## 6. Coloured Cement

- ❖ Uses
  - ☐ For floor finishing works
  - ☐ For plastering
  - ☐ For decorative works
  - ☐ For manufacture of tiles and cast stones

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89

# TYPES OF CEMENT

## 7. Quick Setting Cement

- ❖ Obtained by reducing gypsum during the manufacture of OPC
- ❖ As it sets quickly it is used for
  - ☐ Under water construction
  - ☐ Grouting

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90



# TYPES OF CEMENT

## 8. Expansive Cement

- ❖ OPC shrinks during setting and drying, while expansive cement suffers hardly any change in volume during drying
- ❖ Manufactured by mixing 8 parts of sulpho-aluminate clinker with 100 parts of Portland cement and 15 parts of stabiliser
- ❖ Uses
  - ☐ For machine foundations
  - ☐ Cement Grouting in prestress concrete ducts
  - ☐ Concrete repair works

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91

# TYPES OF CEMENT

## 9. Acid Resisting Cement

- ❖ Resists attack of acids
- ❖ Prepared by adding materials like quartz, sodium silicates to ordinary cement
- ❖ Uses
  - ☐ For the construction of acid resisting floors in factories

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92

# TYPES OF CEMENT

## 10. Sulphate Resisting Cement

- ❖ The percentage of tricalcium aluminate  $< 5\%$  - this increases sulphate resistance
- ❖ Uses
  - ☐ For structures which may be damaged due to severe alkaline conditions like canal lining, culverts and syphon etc.

01-07-2023

93

# USES OF CEMENT

| Sl.no | situations                                              | Type of cement                                                                                           |
|-------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1     | All types of ordinary constructions                     | Ordinary Portland cement                                                                                 |
| 2     | Prefabricated concrete constructions                    | Rapid hardening cement                                                                                   |
| 3     | Road repair works                                       | Rapid hardening cement                                                                                   |
| 4     | For early removal of formworks, cold weather concreting | Rapid hardening cement                                                                                   |
| 5     | Underwater construction, grouting operation             | Quick setting cement                                                                                     |
| 6     | Mass concrete construction (dam, retaining wall)        | <ul style="list-style-type: none"> <li>• Low heat cement</li> <li>• Portland Pozzolana cement</li> </ul> |

01-07-2023

94

# USES OF CEMENT

| Sl.no | situations                           | Type of cement                                                                                                                               |
|-------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 7     | To resist sulphate attack            | Sulphate resisting cement                                                                                                                    |
| 8     | Hot Weather concreting               | Low heat cement                                                                                                                              |
| 9     | Marine conditions                    | <ul style="list-style-type: none"> <li>• Portland Pozzolana cement</li> <li>• High alumina cement</li> <li>• Portland slag cement</li> </ul> |
| 10    | Sewage treatment plants, R.C.C pipes | Sulphahte resisting cement                                                                                                                   |

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95

## CEMENT CONCRETE

- Concrete is an artificial material obtained by hardening of the mixture of

❖ **binding material** (lime, cement etc)

❖ **fine aggregate** (sand)

❖ **coarse aggregate** (crushed rock, gravel etc)

❖ **Water** in predetermined ratios

- Lime concrete – binding material is lime
- Cement concrete – binding material is cement

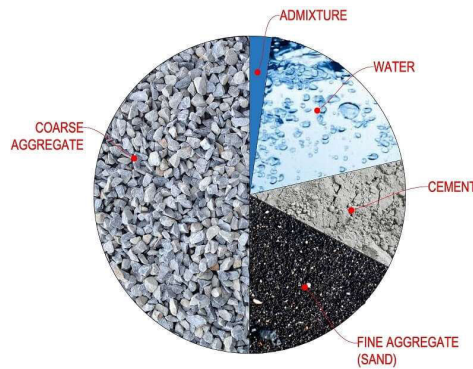


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96

## CEMENT CONCRETE - INGREDIENTS

1. Portland cement
2. Fine aggregate
3. Coarse aggregate
4. Water
5. Admixture



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97

## CEMENT CONCRETE - INGREDIENTS

### 1. Portland cement

- Most important constituent of concrete.
- Majority of the cement concrete work in building construction is done with OPC.
- IS 456-2000 specifies the standard requirements.

### 2. Fine aggregate

- Sand is the principal form.
- Helps to increase the volume of concrete – reduces the cost.
- Inert, strong, hard, sharp and angular.
- IS 373-1970 specifies the standard requirements.

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98

## CEMENT CONCRETE - INGREDIENTS

### 3. Coarse aggregate

- Crushed stones and gravel are the common materials.
- Blast furnace slag, coal ash, coke-breeze etc may be used to for light weight and insulating concrete.
- For most works 20mm aggregate is suitable.
- IS 373-1970 specifies the standard requirements.

### 4. Water

- Used for mixing of constituents of concrete and also for curing.
- Act as lubricant for the fine and coarse aggregate.
- Acts chemically with cement to form the binding paste for the aggregate and reinforcement.

01-07-2023 Potable water are generally considered satisfactory.

99

# CEMENT CONCRETE - INGREDIENTS

## 5. Admixtures

- A material, other than normal constituents (cement, aggregate, water) that is used as an ingredient.
- Added to the batch immediately before or during mixing to
  - ✓ give certain improved qualities
  - ✓ For changing different physical properties in its fresh or hardened stages
- Improve the concrete with respect to strength, hardness, workability or may change setting time.
- Examples – alum, aluminium sulphate, barium oxide, calcium chloride, coal ash, iron oxide, lime, potassium chloride.

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100

# CEMENT CONCRETE - PROPERTIES

## 1. Properties of fresh concrete

- Workability** – property that determines the amount of internal work necessary to produce full compaction (**IS 1199-1959**)
- Segregation**
  - ❖ Separation of coarser particles from the concrete mix
  - ❖ Due to lack of sufficient quantity of finer particles in the mix or dropping or pouring concrete from a greater height
  - ❖ Reduces strength of hardened mass
  - ❖ Minimized by restricting quantity of water, careful handling, transporting, placing and compacting

Segregation in Fresh Concrete



Segregation of Concrete

01-07-2023

101

# CEMENT CONCRETE - PROPERTIES

## 1. Properties of fresh concrete

- Bleeding**
  - ❖ Process of rise of water along with cement particles, to the surface of freshly laid concrete
  - ❖ Happens when there is excessive quantity of water on the mix or when there is excessive compaction
  - ❖ Causes formation of pores on the surface – makes concrete weak
  - ❖ Controlled by proper mix design and using lesser water content

BLEEDING IN CONCRETE



01-07-2023

102

## CEMENT CONCRETE - PROPERTIES

### 2. Properties of hardened concrete

#### a) *Strength*

- ❖ Depends upon grades of concrete, water cement ratio, method of compaction, and time period elapsed
- ❖ Strength of concrete goes on increasing, even after 28 days of casting, though at a slower rate

#### b) *Shrinkage*

- ❖ Concrete shrinks with age
- ❖ Depends upon the constituents of concrete, size of the member and environmental conditions

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## CEMENT CONCRETE - PROPERTIES

### 2. Properties of hardened concrete

#### c) *Creep*

- ❖ Permanent dimensional changes that take place due to loading
- ❖ Depends upon constituents of concrete, size of the member, environmental conditions, stress in the concrete, age of concrete at loading and duration of loading



#### d) *Thermal expansion*

- ❖ Coefficient of thermal expansion depends on nature of cement, nature of aggregate, the relative humidity and size of sections

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104

## CEMENT CONCRETE - PROPERTIES

### 2. Properties of hardened concrete

#### e) *Imperviousness*

- ❖ Resistance of the concrete to the flow of water through the pores.
- ❖ When excess water in concrete evaporates, it leaves voids inside the concrete.
- ❖ May be made impermeable by using low water cement ratio, proper compacting, use of dense and well graded aggregate and continuous curing at low temperature conditions.

01-07-2023

105

## CEMENT CONCRETE - TYPES

1. Plain cement concrete
2. Reinforced cement concrete
3. Fiber reinforced concrete
4. Pre – cast concrete
5. Pre –stressed concrete

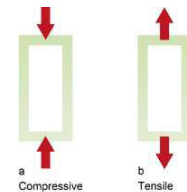
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106

## CEMENT CONCRETE - TYPES

### 1. Plain Cement Concrete (PCC)

- A mixture of **cement, water, fine aggregate, coarse aggregate** and **admixture**.
- The proportion of ingredients depends upon the grade of mix ( $M_{10}$ ,  $M_{15}$ ,  $M_{20}$ ) required for meeting the requirements of the job.
- Used where compressive strength and weight are the principle requirement and where tensile stresses are absent or extremely low such as levelling, bedding for footings, concrete roads.



01-07-2023

107

## CEMENT CONCRETE - TYPES

### 2. Reinforced Cement Concrete (RCC)

- PCC is strong in compression but, its tensile strength is low.
- So PCC is limited to the structures in pure compression.
- Steel is strong in tensile strength and weak in compressive strength.
- Hence the combination of both becomes an ideal solution.
- The composite material of steel and concrete – RCC.
- Steel rods of diameter varying from 5mm to 40mm are used for reinforcing.
- Used as structural element such as footings, columns, beams, roofs and slabs.
- Has long service life, low maintenance and better resistance to fire compared to steel.



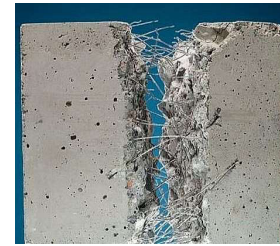
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108

## CEMENT CONCRETE - TYPES

### 3. Fiber Reinforced Concrete (FRC)

- Composite material consisting of mixture of cement mortar or concrete and discontinuous, discrete, uniformly dispersed suitable fibers.
- Fibers may be steel, glass, natural or synthetic.
- They bridge the cracks that develop in concrete and increase ductility of concrete elements.
- They impart resistance to impact, lowers permeability of concrete matrix and reduce the bleeding of water.
- Useful in hydraulic structures, airfield pavements, highways, bridge decks, tunnel lining, heavy duty floors.



01-07-2023

109

## CEMENT CONCRETE - TYPES

### 4. Pre-cast Concrete

- The units or members of various types are cast in separate forms before they are placed in the structure.
- Cast in the building site or at a casting yard; transported to the site by trucks; finally placed in position by cranes or other devices.
- Requires very less scaffolding and form work.
- Concrete with better quality can be produced, as a better quality control at the production yard is possible.



01-07-2023

110

## CEMENT CONCRETE - TYPES

### 5. Pre-stressed Concrete

- Tensioned wires/strands (2,3 or 7 wires)/ tendons (group of wires or strands) are used in place of reinforcing bars.
- The unique characteristic of pre-stressed concrete allow predetermined, engineering stresses to be place in members to counteract stresses that occur when the unit is subjected to service loads.
- Pre-stressing may be achieved by elements outside the concrete (external pre-stressing) or by elements located inside the concrete member (internal pre-stressing) or tension may be applied before casting
- Used to produce beams, floors or bridges with longer spans (>30m)



01-07-2023

111



## STEEL

- Most popular and a versatile construction material widely used for making RCC.
- RCC is a composite material made up of concrete and some form of reinforcement – most commonly steel rods, bars, wires or mesh of steel rods or steel wires.
- Important features such as strength, flexibility and ductility are of great advantage in construction industry.
- The properties of thermal expansion are for both steel and concrete are approximately the same.
- Thermal expansion and excellent bendability and bonding properties makes steel the best material as reinforcement in concrete structures.



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## STEEL

- 2 types – *plain carbon steel* and *alloy steel*
- Plain carbon steel
  - is a metal alloy.
  - It is a combination of two elements, iron and carbon, where other elements are present in quantities too small to affect its properties
  - Carbon content may vary from .01% to 1.5%
  - Increase in carbon content gives strength and hardness to steel, but reduces ductility and malleability
- Based on carbon content, plain carbon steel is further divided into
  1. Mild steel
  2. Medium carbon steel
  3. High carbon steel



01-07-2023

113

## STEEL

1. Mild steel
  - ❖ Percentage of carbon varies from 0.15% to 0.25%
  - ❖ Moderately strong and have good weldability, malleable, ductile and elastic
  - ❖ Mainly used as rolled sections like I-section, T-section, square or round bars
  - ❖ Low tensile strength – not used for RCC
2. Medium carbon steel
  - ❖ Carbon content varies from 0.3% to 0.6%
  - ❖ High strength, but low weldability
3. High carbon steel
  - ❖ Carbon content varies from 0.65% to 1.5%
  - ❖ Gets hard and tough by heat treatment, weldability is poor

01-07-2023

114

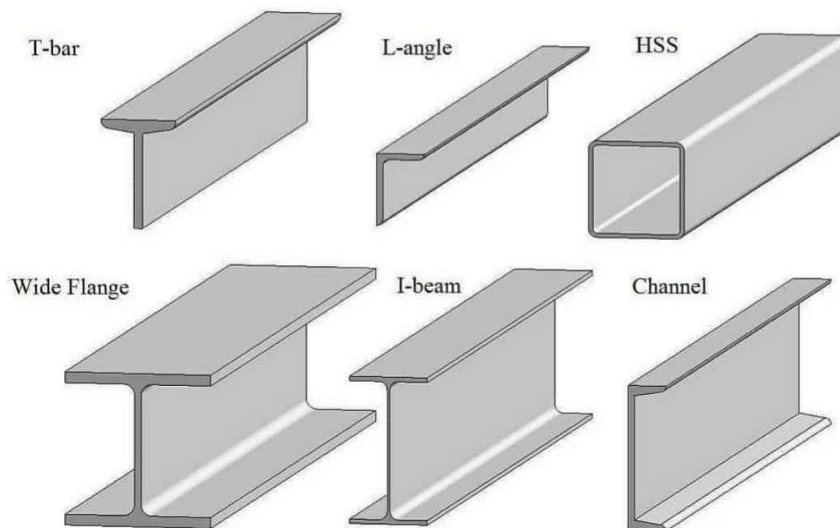


## STEEL – COMMERCIAL FORMS

1. Angle section
2. Channel section
3. I - section
4. T - section
5. Flat sections
6. Corrugated sheets
7. Round bars
8. Square bar
9. Ribbed HYSD bars ( *High Yield Strength Deformed* )

01-07-2023

115



01-07-202

116



**Rolled steel tubes**



**Rolled steel flats**

01-07-2023

117



**Corrugated sheets**



**Ribbed bars**

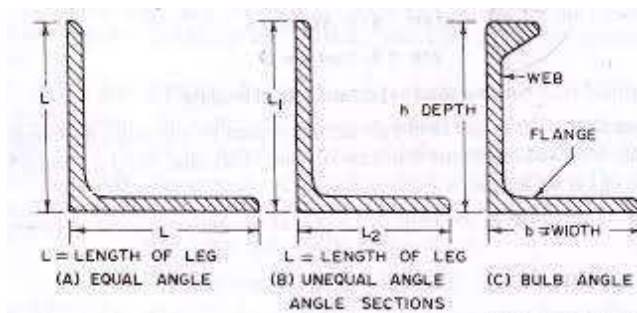
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118

## STEEL – COMMERCIAL FORMS

### 1. Angle section

- Consist of two legs – equal or unequal length
- Available is size varying from 20 x 20 x 3mm to 200 x 200 x 25mm
- Indian Standard bulb angle – angle with bulged portion at the end of the leg



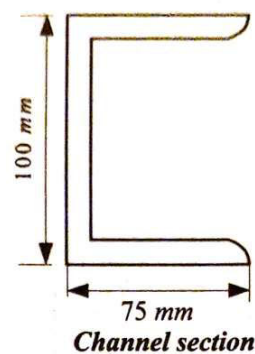
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119

## STEEL – COMMERCIAL FORMS

### 2. Channel section

- Consist of a web with two equal flanges
- Designated by height of web & width of flange
- Available in sizes from 100 x 45 mm to 400 x 100 mm



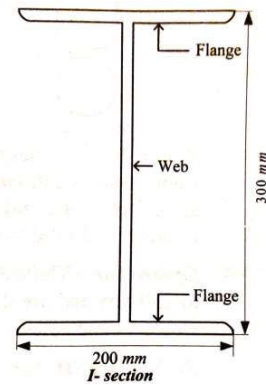
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120

## STEEL – COMMERCIAL FORMS

### 3. I-section

- Consist of two flanges connected by a web.
- Designated by overall depth, width of flange and weight/m length.
- Available sizes are 75 x 50 mm to 600 x 210 mm of varying thickness.
- Used as lintels, beams, columns, members of steel frameworks, built up columns for carrying lateral loads.



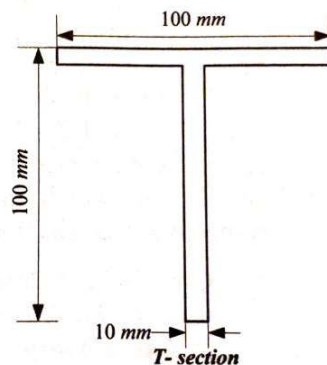
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121

## STEEL – COMMERCIAL FORMS

### 4. T-section

- Consist of a flange & a web.
- Designated by overall dimensions and thickness.
- Available in sizes from 20 x 20 x 3 mm to 150 x 150 x 10mm.
- Used as member of the steel roof trusses and to form built-up sections.



01-07-2023

122

## STEEL – COMMERCIAL FORMS

### 5. Flat section

- Plate like sections with more length and less width.
- Available from 3 mm to 40mm thickness and 10mm to 400mm width.
- Used in the fabrication of grill work for windows and gates, built up beams and columns, plate girder bridges etc.

### 6. Corrugated section

- Formed by passing steel sheets through grooves.
- The grooves bend and press steel sheets and corrugations are formed.
- Are galvanized and are called galvanized iron sheets or G.I sheets – used for roof covering and cladding.

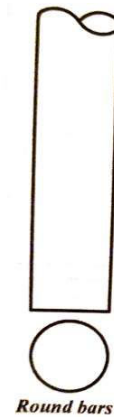
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123

## STEEL – COMMERCIAL FORMS

### 7. Round bars

- Available in circular cross section with diameter varying from 5mm to 250mm.
- Both solid and hollow sections are available.
- Hollow sections are called pipes and are denoted by *Indian Standard Round Bar (ISRO)*.



01-07-2023

124

## STEEL – COMMERCIAL FORMS

### 8. Square bars

- Available in square section with sides varying from 5mm to 250mm and are denoted *ISSQ*.
- Widely used in steel grill works for windows and gates.



01-07-2023

125

## STEEL – COMMERCIAL FORMS

### 9. Ribbed HYSD bars

- Are deformed high strength steel.
- Have ribs or projections on the surface which increases the bond between concrete and steel bars.
- Mechanically treated at high temperature and rolled and twisted for shaping.
- Have more tensile strength than mild steel ( $>400\text{Mpa}$ ).
- Have satisfactory malleability and weldability.
- 2 types - TOR and TMT – high strength steel bars.
- TOR refers to cold treated bars and TMT to hot treated bars and are graded as Fe415 and Fe500 i.e, yield strength is  $415\text{ N/mm}^2$  and  $500\text{ N/mm}^2$  respectively.



01-07-2023

126

# SYLLABUS

- Modern construction materials (brief discussion only)
  - Architectural glass,
  - Ceramics
  - plastics
  - composite materials
  - thermal and acoustic insulating materials
  - decorative panels
  - waterproofing materials
  - modern uses of gypsum,
  - pre-fabricated building components